Database Schema: Order Management System

Enterprise-grade PostgreSQL schema designed for Spring Boot (JPA), Angular, and JasperReports, with comprehensive auditing via `updated_at` columns.

1. categories (Item Categorization)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PRIMARY KEY	Unique identifier
name	VARCHAR(100)	NOT NULL	e.g., Starters, Main Course
is_active	BOOLEAN	DEFAULT TRUE	Soft delete flag
created_at	TIMESTAMP	NOT NULL	JPA @CreatedDate
updated_at	TIMESTAMP	NOT NULL	JPA @LastModifiedDate

2. items (Menu Management)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PRIMARY KEY	Unique identifier
category_id	UUID	FOREIGN KEY	Links to categories
name	VARCHAR(150)	NOT NULL	Item name
price	DECIMAL(10,2)	NOT NULL	Current selling price
is_available	BOOLEAN	DEFAULT TRUE	Out-of-stock toggle
created_at	TIMESTAMP	NOT NULL	Creation timestamp
updated_at	TIMESTAMP	NOT NULL	Last update timestamp

3. restaurant_tables (QR & Dine-in Management)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PRIMARY KEY	Unique identifier
table_number	VARCHAR(20)	UNIQUE, NOT NULL	e.g., "T-01"
qr_token	UUID	UNIQUE, NOT NULL	Secure token for QR code scanning
status	VARCHAR(50)	DEFAULT 'AVAILABLE'	AVAILABLE, OCCUPIED, RESERVED
created_at	TIMESTAMP	NOT NULL	Creation timestamp
updated_at	TIMESTAMP	NOT NULL	Last update timestamp

4. orders (The Core Transaction Hub)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PRIMARY KEY	Unique identifier
order_number	VARCHAR(50)	UNIQUE, NOT NULL	Human-readable sequence
order_type	VARCHAR(20)	NOT NULL	DINE_IN, TAKE_AWAY
status	VARCHAR(20)	NOT NULL	SOFT_ORDER, CONFIRMED, BILLED
table_id	UUID	FK, NULLABLE	Null for Take-Away
sub_total	DECIMAL(10,2)	NOT NULL	Pre-tax/discount sum
tax_amount	DECIMAL(10,2)	DEFAULT 0.00	Calculated tax
discount_amount	DECIMAL(10,2)	DEFAULT 0.00	Applied discounts
grand_total	DECIMAL(10,2)	NOT NULL	Final payable amount
created_at	TIMESTAMP	NOT NULL	Order start time
updated_at	TIMESTAMP	NOT NULL	Order modification time

5. order_items (Details & Fast-Moving Tracking)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PRIMARY KEY	Unique identifier
order_id	UUID	FOREIGN KEY	Links to orders
item_id	UUID	FOREIGN KEY	Links to items
quantity	INTEGER	NOT NULL	Number of units ordered
unit_price	DECIMAL(10,2)	NOT NULL	Captured at order time
total_price	DECIMAL(10,2)	NOT NULL	quantity * unit_price
notes	VARCHAR(255)	NULLABLE	Customer requests
created_at	TIMESTAMP	NOT NULL	Row creation time
updated_at	TIMESTAMP	NOT NULL	Row update time

6. payments (Billing & Receipts)

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	UUID	PRIMARY KEY	Unique identifier
order_id	UUID	FOREIGN KEY	Links to orders
amount_paid	DECIMAL(10,2)	NOT NULL	Supports split payments
payment_method	VARCHAR(50)	NOT NULL	CASH, CARD, UPI
payment_status	VARCHAR(20)	NOT NULL	PENDING, SUCCESS, FAILED
created_at	TIMESTAMP	NOT NULL	Payment initiation time
updated_at	TIMESTAMP	NOT NULL	Payment resolution time

Design Rationale

Auditing Compatibility: The explicit inclusion of `created_at` and `updated_at` across all tables perfectly complements Spring Data JPA's `@EnableJpaAuditing`. By mapping these to `@CreatedDate` and `@LastModifiedDate` in a mapped superclass, the application automatically handles lifecycle timestamps without manual repository updates.

Immutability & Analytics: Storing historical prices directly in the `order_items` table ensures JasperReports generates consistent past receipts even if catalog prices change. Furthermore, standardizing audit timestamps simplifies aggregate functions for day/month/year performance dashboards.